

## Cardinality rationale for functions

**Key idea for cardinality:** Counting distinct elements is a way of labelling elements with natural numbers. This is a function! In general, functions let us associate elements of one set with those of another. If the association is “*good*”, we get a correspondence between the elements of the subsets which can relate the sizes of the sets.

## Musical chairs analogy

*Analogy:* Musical chairs



People try to sit down when the music stops

Person☼ sits in Chair 1, Person☺ sits in Chair 2,

Person☹ is left standing!

What does this say about the number of chairs and the number of people?

## Injective functions visually

Informally, a function being one-to-one means “no duplicate images”.

## Cardinality lower bound definition

**Definition:** For nonempty sets  $A, B$ , we say that **the cardinality of  $A$  is no bigger than the cardinality of  $B$** , and write  $|A| \leq |B|$ , to mean there is a one-to-one function with domain  $A$  and codomain  $B$ . Also, we define  $|\emptyset| \leq |B|$  for all sets  $B$ .

## Injective cardinality musical chairs

*In the analogy:* The function  $sitter : \{Chair1, Chair2\} \rightarrow \{Person\otimes, Person\ominus, Person\odot\}$  given by  $sitter(Chair1) = Person\otimes$ ,  $sitter(Chair2) = Person\ominus$ , is one-to-one and witnesses that

$$|\{Chair1, Chair2\}| \leq |\{Person\otimes, Person\ominus, Person\odot\}|$$

## Cardinality upper bound definition

**Definition:** For nonempty sets  $A, B$ , we say that **the cardinality of  $A$  is no smaller than the cardinality of  $B$** , and write  $|A| \geq |B|$ , to mean there is an onto function with domain  $A$  and codomain  $B$ . Also, we define  $|A| \geq |\emptyset|$  for all sets  $A$ .

## Surjective cardinality musical chairs

*In the analogy:* The function  $triedToSit : \{Person\otimes, Person\ominus, Person\odot\} \rightarrow \{Chair1, Chair2\}$  given by  $triedToSit(Person\otimes) = Chair1$ ,  $triedToSit(Person\ominus) = Chair2$ ,  $triedToSit(Person\odot) = Chair2$ , is onto and witnesses that

$$|\{Person\otimes, Person\ominus, Person\odot\}| \geq |\{Chair1, Chair2\}|$$

## Cardinality properties

### Properties of cardinality

$$\begin{aligned} & \forall A ( |A| = |A| ) \\ & \forall A \forall B ( |A| = |B| \rightarrow |B| = |A| ) \\ & \forall A \forall B \forall C ( (|A| = |B| \wedge |B| = |C|) \rightarrow |A| = |C| ) \end{aligned}$$

*Extra practice with proofs:* Use the definitions of bijections to prove these properties.

# Cardinality power sets

*Recall:* When  $U$  is a set,  $\mathcal{P}(U) = \{X \mid X \subseteq U\}$

*Key idea:* For finite sets, the power set of a set has strictly greater size than the set itself. Does this extend to infinite sets?

**Definition:** For two sets  $A, B$ , we use the notation  $|A| < |B|$  to denote  $(|A| \leq |B|) \wedge \neg(|A| = |B|)$ .

|                    |                                                                 |                                          |
|--------------------|-----------------------------------------------------------------|------------------------------------------|
| $\emptyset = \{\}$ | $\mathcal{P}(\emptyset) = \{\emptyset\}$                        | $ \emptyset  <  \mathcal{P}(\emptyset) $ |
| $\{1\}$            | $\mathcal{P}(\{1\}) = \{\emptyset, \{1\}\}$                     | $ \{1\}  <  \mathcal{P}(\{1\}) $         |
| $\{1, 2\}$         | $\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$ | $ \{1, 2\}  <  \mathcal{P}(\{1, 2\}) $   |

## $\mathbb{N}$ and its power set

Example elements of  $\mathbb{N}$

Example elements of  $\mathcal{P}(\mathbb{N})$

**Claim:**  $|\mathbb{N}| \leq |\mathcal{P}(\mathbb{N})|$

**Claim:** There is an uncountable set. Example: \_\_\_\_\_

**Proof:** By definition of countable, since \_\_\_\_\_ is not finite, **to show** is  $|\mathbb{N}| \neq |\mathcal{P}(\mathbb{N})|$ .

Rewriting using the definition of cardinality, **to show** is

Towards a proof by universal generalization, consider an arbitrary function  $f : \mathbb{N} \rightarrow \mathcal{P}(\mathbb{N})$ .

**To show:**  $f$  is not a bijection. It's enough to show that  $f$  is not onto.

Rewriting using the definition of onto, **to show:**

$$\neg \forall B \in \mathcal{P}(\mathbb{N}) \exists a \in \mathbb{N} ( f(a) = B )$$

. By logical equivalence, we can write this as an existential statement:

---

In search of a witness, define the following collection of nonnegative integers:

$$D_f = \{n \in \mathbb{N} \mid n \notin f(n)\}$$

. By definition of power set, since all elements of  $D_f$  are in  $\mathbb{N}$ ,  $D_f \in \mathcal{P}(\mathbb{N})$ . It's enough to prove the following Lemma:

**Lemma:**  $\forall a \in \mathbb{N} ( f(a) \neq D_f )$ .

**Proof of lemma:**

By the Lemma, we have proved that  $f$  is not onto, and since  $f$  was arbitrary, there are no onto functions from  $\mathbb{N}$  to  $\mathcal{P}(\mathbb{N})$ . QED

**Where does  $D_f$  come from?** The idea is to build a set that would “disagree” with each of the images of  $f$  about some element.

| $n \in \mathbb{N}$ | $f(n) = X_n$ | Is $0 \in X_n$ ?    | Is $1 \in X_n$ ?    | Is $2 \in X_n$ ?    | Is $3 \in X_n$ ?    | Is $4 \in X_n$ ?    | ... | Is $n \in D_f$ ?    |
|--------------------|--------------|---------------------|---------------------|---------------------|---------------------|---------------------|-----|---------------------|
| 0                  | $f(0) = X_0$ | <b>Y</b> / <b>N</b> | Y / N               | Y / N               | Y / N               | Y / N               | ... | <b>N</b> / <b>Y</b> |
| 1                  | $f(1) = X_1$ | Y / N               | <b>Y</b> / <b>N</b> | Y / N               | Y / N               | Y / N               | ... | <b>N</b> / <b>Y</b> |
| 2                  | $f(2) = X_2$ | Y / N               | Y / N               | <b>Y</b> / <b>N</b> | Y / N               | Y / N               | ... | <b>N</b> / <b>Y</b> |
| 3                  | $f(3) = X_3$ | Y / N               | Y / N               | Y / N               | <b>Y</b> / <b>N</b> | Y / N               | ... | <b>N</b> / <b>Y</b> |
| 4                  | $f(4) = X_4$ | Y / N               | Y / N               | Y / N               | Y / N               | <b>Y</b> / <b>N</b> | ... | <b>N</b> / <b>Y</b> |
| ⋮                  |              |                     |                     |                     |                     |                     |     |                     |

# Cardinality rationals reals

## Comparing $\mathbb{Q}$ and $\mathbb{R}$

Both  $\mathbb{Q}$  and  $\mathbb{R}$  have no greatest element.

Both  $\mathbb{Q}$  and  $\mathbb{R}$  have no least element.

The quantified statement

$$\forall x \forall y (x < y \rightarrow \exists z (x < z < y))$$

is true about both  $\mathbb{Q}$  and  $\mathbb{R}$ .

Both  $\mathbb{Q}$  and  $\mathbb{R}$  are infinite. But,  $\mathbb{Q}$  is countably infinite whereas  $\mathbb{R}$  is uncountable.

## The set of real numbers

$$\mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R}$$

**Order axioms** (Rosen Appendix 1):

|              |                                                                                                                                    |
|--------------|------------------------------------------------------------------------------------------------------------------------------------|
| Reflexivity  | $\forall a \in \mathbb{R} (a \leq a)$                                                                                              |
| Antisymmetry | $\forall a \in \mathbb{R} \forall b \in \mathbb{R} ( (a \leq b \wedge b \leq a) \rightarrow (a = b) )$                             |
| Transitivity | $\forall a \in \mathbb{R} \forall b \in \mathbb{R} \forall c \in \mathbb{R} ( (a \leq b \wedge b \leq c) \rightarrow (a \leq c) )$ |
| Trichotomy   | $\forall a \in \mathbb{R} \forall b \in \mathbb{R} ( (a = b \vee b > a \vee a < b) )$                                              |

**Completeness axioms** (Rosen Appendix 1):

|                   |                                                                                                                                                                                            |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Least upper bound | Every nonempty set of real numbers that is bounded above has a least upper bound                                                                                                           |
| Nested intervals  | For each sequence of intervals $[a_n, b_n]$ where, for each $n$ , $a_n < a_{n+1} < b_{n+1} < b_n$ , there is at least one real number $x$ such that, for all $n$ , $a_n \leq x \leq b_n$ . |

Each real number  $r \in \mathbb{R}$  is described by a function to give better and better approximations

$$x_r : \mathbb{Z}^+ \rightarrow \{0, 1\} \quad \text{where } x_r(n) = n^{\text{th}} \text{ bit in binary expansion of } r$$

| $r$                             | Binary expansion | $x_r$                                                                                                                                                                                                                                                                          |
|---------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.1                             | 0.00011001...    | $x_{0.1}(n) = \begin{cases} 0 & \text{if } n > 1 \text{ and } (n \bmod 4) = 2 \\ 0 & \text{if } n = 1 \text{ or if } n > 1 \text{ and } (n \bmod 4) = 3 \\ 1 & \text{if } n > 1 \text{ and } (n \bmod 4) = 0 \\ 1 & \text{if } n > 1 \text{ and } (n \bmod 4) = 1 \end{cases}$ |
| $\sqrt{2} - 1 = 0.4142135\dots$ | 0.01101010...    | Use linear approximations (tangent lines from calculus) to get algorithm for bounding error of successive operations. Define $x_{\sqrt{2}-1}(n)$ to be $n^{\text{th}}$ bit in approximation that has error less than $2^{-(n+1)}$ .                                            |

**Claim:**  $\{r \in \mathbb{R} \mid 0 \leq r \wedge r \leq 1\}$  is uncountable.

*Approach 1:* Mimic proof that  $\mathcal{P}(\mathbb{Z}^+)$  is uncountable.

**Proof:** By definition of countable, since  $\{r \in \mathbb{R} \mid 0 \leq r \wedge r \leq 1\}$  is not finite, **to show** is  $|\mathbb{N}| \neq |\{r \in \mathbb{R} \mid 0 \leq r \wedge r \leq 1\}|$ .

**To show** is  $\forall f : \mathbb{Z}^+ \rightarrow \{r \in \mathbb{R} \mid 0 \leq r \wedge r \leq 1\}$  ( $f$  is not a bijection) . Towards a proof by universal generalization, consider an arbitrary function  $f : \mathbb{Z}^+ \rightarrow \{r \in \mathbb{R} \mid 0 \leq r \wedge r \leq 1\}$ . **To show:**  $f$  is not a bijection. It's enough to show that  $f$  is not onto. Rewriting using the definition of onto, **to show:**

$$\exists x \in \{r \in \mathbb{R} \mid 0 \leq r \wedge r \leq 1\} \forall a \in \mathbb{N} ( f(a) \neq x )$$

In search of a witness, define the following real number by defining its binary expansion

$$d_f = 0.b_1b_2b_3 \dots$$

where  $b_i = 1 - b_{ii}$  where  $b_{jk}$  is the coefficient of  $2^{-k}$  in the binary expansion of  $f(j)$ . Since<sup>1</sup>  $d_f \neq f(a)$  for any positive integer  $a$ ,  $f$  is not onto.

*Approach 2:* Nested closed interval property

**To show**  $f : \mathbb{N} \rightarrow \{r \in \mathbb{R} \mid 0 \leq r \wedge r \leq 1\}$  is not onto. **Strategy:** Build a sequence of nested closed intervals that each avoid some  $f(n)$ . Then the real number that is in all of the intervals can't be  $f(n)$  for any  $n$ . Hence,  $f$  is not onto.

Consider the function  $f : \mathbb{N} \rightarrow \{r \in \mathbb{R} \mid 0 \leq r \wedge r \leq 1\}$  with  $f(n) = \frac{1+\sin(n)}{2}$

| $n$      | $f(n)$      | Interval that avoids $f(n)$ |
|----------|-------------|-----------------------------|
| 0        | 0.5         |                             |
| 1        | 0.920735... |                             |
| 2        | 0.954649... |                             |
| 3        | 0.570560... |                             |
| 4        | 0.121599... |                             |
| $\vdots$ |             |                             |

---

<sup>1</sup>There's a subtle imprecision in this part of the proof as presented, but it can be fixed.